

# Moritz Grauer

Graphics Programmer · Game Developer · Researcher

## CONTACT

-  [vexury.dev@gmail.com](mailto:vexury.dev@gmail.com)
-  [github.com/Vexury](https://github.com/Vexury)
-  [vexury.itch.io](https://vexury.itch.io)
-  Karlsruhe, Germany

## SKILLS

### LANGUAGES

- C++C#JavaPython
- GLSLHLSL

### GRAPHICS APIS

- VulkanOpenGL

### RENDERING

- Path TracingBVH / SAH
- PBR / GGX
- Global Illumination
- NEE / MIS

### TOOLS

- GitLaTeXCPU Profiling
- GPU Profiling

### LANGUAGES

- GermanNative
- EnglishFluent

## ABOUT

I build rendering systems and research ray tracing, with a focus on GPU acceleration structures and global illumination. Four years as a researcher and teaching associate at KIT, supervising nine theses and teaching courses on photorealistic rendering and computer graphics. Outside of research I participate in game jams and develop my own path-tracing engine.

## EXPERIENCE

- Computer Graphics Researcher @ KIT**  
Feb 2022 - Apr 2026 · Karlsruhe  
Research on ray tracing acceleration structures (BVH with binned SAH), GPU rendering pipelines, and global illumination. Teaching associate for Photorealistic Image Synthesis, Computer Graphics, and Computer Graphics 2. Supervised 9 theses on topics including wide BVH traversal on GPU, mesh shader volume rendering, and neural BVH construction.
- QA Engineer @ Gameforge**  
Oct 2020 - Sep 2021 · Karlsruhe  
Quality assurance for live online games. Bug tracking, test case documentation, and cross-team coordination for release validation.
- Programming Tutor @ KIT**  
Oct 2016 - Jul 2019 · Karlsruhe  
Led exercise sessions for undergraduate programming courses. Graded assignments and held office hours.

## EDUCATION

- M.Sc. Informatics**  
KIT · 2018 - 2021  
Focus on computer graphics and high-performance computing. Exchange semester at Chalmers University of Technology, Gothenburg.
- B.Sc. Informatics**  
KIT · 2015 - 2018  

★ **Studienstiftung des deutschen Volkes scholarship**

## TEACHING

|                                                                |                                           |
|----------------------------------------------------------------|-------------------------------------------|
| <b>Photorealistic Image Synthesis</b><br>S'22 · S'23 · W'24/25 | <b>Computer Graphics 2</b><br>S'24 · S'25 |
| <b>Computer Graphics</b><br>W'22/23 · W'25/26                  | <b>Lab Course Supervision</b><br>W'23/24  |

## SELECTED PROJECTS

- VexEngine** [github ↗](#) [project page ↗](#)  
Custom C++ rendering engine with dual OpenGL/Vulkan backends. Progressive CPU and GPU path tracing with hardware ray tracing, Cook-Torrance GGX PBR materials, BVH with binned SAH, NEE and MIS light sampling, post-processing pipeline (tonemapping, bloom), and a full dockable editor UI.
- Necrocure** [itch.io ↗](#)  
Mini Jame Gam #55 entry. Alien abduction action minigame built in 58 hours. Handled all programming, VFX, and UI; created with one collaborator handling art and audio.
- Quack 3** [ggj ↗](#)  
Global Game Jam 2026 entry. Fast-paced FPS arena shooter developed over 48 hours in a team of three, with all art and audio created on-site during the jam.